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Project Title:	Investigation into the Quantification and Control of Steady and Unsteady Flow Phenomena in Passive Gas-Liquid Cyclonic Separators		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan (Case Western Reserve University)		
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Task Description:

Phase separators are devices designed to divide flows of intermingling gas and liquid into separate streams that are individually purely liquid or purely gas. Separation of the respective phases is of considerable interest to NASA as many spacecraft systems can only function effectively if the flow they receive is composed exclusively of the one phase that they are designed to accommodate. While separation occurs naturally on the Earth, the absence of buoyancy in a microgravity environment allows the undesired mixtures of gas and liquid to remain combined unopposed. Due to this reluctance of the phases to separate, installation of some type of separation device becomes supremely important for many of the fluid systems that are used by NASA. Cyclonic separators provide the gas-liquid separatory action by swirling the multiphase flow – causing the gas to accumulate along the axis of the vortex as the denser liquid is forced to the walls – thereby allowing segregated extraction of the respective phases. Passive cyclonic separators utilize only the inertia of the incoming flow to accomplish this task and in doing so can avoid the moving parts and reliability concerns of their active brethren. While devices of this type have been successfully employed in experimental settings, more complete investigations and analyses are required before these devices can ever be seriously considered for use in operational space hardware.
The objective of this study is to facilitate the maturation of passive cyclonic separation technology through exhaustive research of a newly designed gas-liquid phase separator evolved from NASA Glenn Research Center's (GRC) existing Cascade Cyclonic Separation Device (CSD-C). Combined experimental and numerical analyses will seek to quantitatively assess and delimit the range of operability of the new separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct – features that have yet to be completely explored in any such device.

The approach is first to assess the range of operability of the new separator design through the careful construction of detailed flow separation maps using data gathered from reduced-gravity flight experimentation in conjunction with ancillary computational studies. Second, the transient and stability behavior of the device will be observed and analyzed. Special attention will be paid both to device-level conduct and also to behaviors that could manifest dysfunction elsewhere at the complete system level. A multiscale hybrid of the volume-of-fluid (VOF) and Euler methods has been coupled to 1-D system-level multiphase solvers in order to permit discernment of these flow features. Lastly, with the ensuing understanding of the causes of any performance-affecting responses, new ways to control transient and unstable operations will be sought; in particular, the insight acquired should facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CMFD as a reliable design tool to further assist in the creation and analysis of new designs.

The combination of the research activity included in this grant with the complementary activity of Dynaflow Inc. of Jessup, MD (Grant NNX09AI35G) has evolved into the Two-Phase Flow Separator Experiment (TPFSE). TPFSE represents a comprehensive investigation of passive cyclone separation technology – an investigation made possible by capabilities that only NASA and the International Space Station can offer.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Phase separation has been identified as a critical mission-enabling capability for spacecraft systems ranging from power conversion to Environmental Control and Life Support Systems. As these systems have little tolerance for failure, extensive investigations and analyses are required before these devices can ever be seriously considered for employment in operational space hardware. Due to the advantages associated with passive vortex devices, cyclonic separators are also finding increasing use in terrestrial applications, such as in the petroleum industry where separators of this type are considered to be an emerging technology. The detailed study of the features of the complicated vortex-driven flow along with the modeling techniques under development for the microgravity investigations could equally be exploited for the advancement of these more Earthly applications.

Task Progress:

The TPFSE (Two-Phase Flow Separator Experiment) project at NASA is designed to help to close these passive cyclonic separation and multiphase systems knowledge gaps in order to start to afford designers the latitude to freely leverage the advantages of multiphase systems in the many scenarios where they arise – helping to enable transformational space exploration and scientific discovery.
Combined experimental, numerical, and phenomenological modeling analyses had been performed for the passive separator designed at CWRU to qualitatively describe its performance throughout the entire project funding period. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities.

The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations. In general the results are promising and demonstrate the benefits of using passive cyclonic separation. However, the need for long-duration microgravity experiments is clearly indicated.

Ground-based and Reduced-gravity Experiment

Both ground-based and reduced-gravity testing has been conducted at CWRU and NASA Glenn Research Center Microgravity Branch. Various diagnostic techniques, including electrical resistance tomography (ERT), pulsed Doppler ultrasonic velocimetry (DPUV) and high speed imaging etc., were used to classify the performance of the separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. had all been studied.

Computational Fluid Dynamics

Computational fluid dynamics has also been heavily pursued by the PI team. The nature of the multiphase flow in microgravity gas-liquid separators presents a number of challenges for CFD simulations. These separators simultaneously possess gas in the form of disperse bubbles and in the form of a continuous gas core. A hybridization of a VOF model with a multifluid Eulerian-Eulerian treatment for dispersed phase has been adopted in order to capture all

	of the pertinent flow physics. The CFD techniques were then used to simulate the separator performance under the conditions of steady and unsteady injection of both single-phase and multiphase flow. Analytical Modeling A phenomenological description of the separator was pursued by the PI team. This can be formalized in a simple control volume analysis that is able to provide insight into the scaling parameters at play governing the device dynamics. The simple system of equations that results can be solved far more quickly than the full CFD modeling approach. TPFSE In May 2013 we had Science Concept Review (SCR) at NASA/GRC based on the work performed at CWRU. We passed the review and the experiment moved into the design phase. The proposed ISS experiment is tentatively scheduled for 2019 after a series of reviews. We have collaborated with NASA/GRC engineers to prepare and design the breadboard experiment system in past year. NOTE that Flight Experiment follow-on is "Cyclonic Two-Phase Flow Separator Experiment in Microgravity," grant NNX14AJ74G, with same principal investigator.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Abstracts for Journals and Proceedings	Kang MF, Hoyt NC, Lee KL, Kharraz AO, Kadambi J, Kamotani Y. "Study of Gas Core Behavior of Passive Cyclonic Two-Phase Separator for Microgravity Applications." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 54. , Nov-2013
Articles in Peer-reviewed Journals	Hoyt NC, Kang MF, Lee KL, Kharraz A, Kadambi J, Kamotani Y. "Study of steady and dynamic behavior of gas core of passive cyclonic separator for space applications." Microgravity Science and Technology. 2013 Oct;25(3):187-200. http://dx.doi.org/10.1007/s12217-013-9346-9 , Oct-2013
Articles in Peer-reviewed Journals	Kang MF, Hoyt NC, Kadambi J, Kamotani Y. "Study of gas core behavior of passive cyclonic two-phase separator for microgravity applications." Microgravity Science and Technology. 2014 Oct;26(3):147-57. http://dx.doi.org/10.1007/s12217-014-9389-6 , Oct-2014